Lecture 12

Verma modules and irreducible highest weights

Objectives. *For each $\lambda \in \mathfrak{h}^*$ we build a universal module $M(\lambda)$ generated by a highest vector. It has a unique simple quotient $L(\lambda)$. Every finite-dimensional simple module is isomorphic to some $L(\lambda)$, with λ dominant integral.*

By the end of this lecture you will be able to:

- Construct $M(\lambda)$ as an induced module.
- Find the weight spaces of $M(\lambda)$ from PBW.
- Prove that each highest-weight module has a unique simple quotient.
- Show that a finite-dimensional simple module is isomorphic to $L(\lambda)$ for a unique $\lambda \in P_+$.

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, positive roots R_+ and simple roots $\alpha_1, \dots, \alpha_r$. Write

$$\mathfrak{n}_\pm = \bigoplus_{\alpha \in R_+} \mathfrak{g}_{\pm\alpha}, \qquad Q_+ = \sum_{i=1}^{r} \mathbb{Z}_{\geq 0}\, \alpha_i.$$

For $\lambda, \mu \in \mathfrak{h}^*$, write $\mu \leq \lambda$ if $\lambda - \mu \in Q_+$.

Departure from the standing convention. Modules in this lecture may be infinite-dimensional.

We import the following results.

(a) Lecture 7 and Lecture 8: $\mathfrak{h}$ is abelian, and $[h, x] = \alpha(h)x$ for $h \in \mathfrak{h}$ and $x \in \mathfrak{g}_\alpha$. Root spaces satisfy $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$ and are one-dimensional. Each simple root α_i has an $\mathfrak{sl}_2$-triple (e_i, h_i, f_i) with $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = \alpha_i^\vee \in \mathfrak{h}$ and

$$[h_i, e_i] = 2e_i, \qquad [h_i, f_i] = -2f_i, \qquad [e_i, f_i] = h_i.$$

(b) Lecture 9: the simple roots are linearly independent, and the coroots $h_1, \dots, h_r$ form a basis of $\mathfrak{h}$. The positive roots are $R_+ = R \cap Q_+$, and $R = R_+ \sqcup (-R_+)$. Independence gives $Q_+ \cap (-Q_+) = 0$, so $\leq$ is a partial order. The dominant integral weights are

$$P_+ = \{\lambda \in \mathfrak{h}^* : \lambda(h_i) \in \mathbb{Z}_{\geq 0} \text{ for } i = 1, \dots, r\} = \sum_{i=1}^{r} \mathbb{Z}_{\geq 0}\, \omega_i,$$

where $\omega_i(h_j) = \delta_{ij}$.

(c) Lecture 11: $\mathfrak{g}$-modules are the same as unital $U(\mathfrak{g})$-modules. For a subalgebra $\mathfrak{a}$, the map $U(\mathfrak{a}) \to U(\mathfrak{g})$ is injective, and ordered PBW monomials in a basis of $\mathfrak{a}$ form a basis of $U(\mathfrak{a})$. For the subalgebras $\mathfrak{n}_\pm$ and $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$, multiplication is a linear isomorphism

$$U(\mathfrak{n}_-) \otimes U(\mathfrak{b}) \longrightarrow U(\mathfrak{g}), \qquad y \otimes b \longmapsto yb.$$

12.1 The induced module

For a $\mathfrak{g}$-module V and $\mu \in \mathfrak{h}^*$, the *weight space* is

$$V_\mu = \{v \in V : hv = \mu(h)v \text{ for all } h \in \mathfrak{h}\}.$$

If $V_\mu \neq 0$, then μ is a *weight* of V. For $h \in \mathfrak{h}$, $x \in \mathfrak{g}_\alpha$ and $v \in V_\mu$, the module axiom and import (a) give

$$h(xv) = x(hv) + [h,x]v = \mu(h)\,xv + \alpha(h)\,xv = (\mu+\alpha)(h)\,xv.$$

Hence

$$\mathfrak{g}_\alpha V_\mu \subseteq V_{\mu+\alpha}.$$

Lemma 12.1 (Weight components). *Let V be a $\mathfrak{g}$-module and*

$$w = w_1 + \cdots + w_k, \qquad w_j \in V_{\mu_j},$$

with distinct $\mu_1, \dots, \mu_k$. Every subspace $W \subseteq V$ that contains w and is stable under $\mathfrak{h}$ contains each w_j. In particular, the sum of the weight spaces of V is direct.

Proof. Fix i. For each $j \neq i$ choose $t_j \in \mathfrak{h}$ with $\mu_i(t_j) \neq \mu_j(t_j)$. Put

$$P_i = \prod_{j \neq i} \frac{t_j - \mu_j(t_j)}{\mu_i(t_j) - \mu_j(t_j)}.$$

The factors commute and act on each V_{μ_l} by scalars. The factor with index j kills w_j, and every factor fixes w_i. Thus

$$P_i w_j = 0 \quad (j \neq i), \qquad P_i w_i = w_i, \qquad P_i w = w_i.$$

Since W is stable under $\mathfrak{h}$, it is stable under P_i. Hence $w_i \in W$.

For directness, take $w = 0$ and $W = 0$. Then every $w_j = 0$. ∎

The subspace $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$ is the *Borel subalgebra*. For $\alpha, \beta \in R_+$, import (a) gives

$$[\mathfrak{h},\mathfrak{h}] = 0, \qquad [\mathfrak{h},\mathfrak{g}_\alpha] \subseteq \mathfrak{g}_\alpha, \qquad [\mathfrak{g}_\alpha,\mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

Then $\alpha+\beta \in Q_+$, and $\alpha+\beta \neq 0$ because $Q_+ \cap (-Q_+) = 0$ by import (b). If $\alpha+\beta$ is a root, it lies in $R \cap Q_+ = R_+$ by import (b). Otherwise $\mathfrak{g}_{\alpha+\beta} = 0$. So $\mathfrak{b}$ is a subalgebra, and $[\mathfrak{b},\mathfrak{b}] \subseteq \mathfrak{n}_+$.

Definition 12.2 (The character $\mathbb{C}_\lambda$). For $\lambda \in \mathfrak{h}^*$, let $\mathbb{C}_\lambda$ be the line $\mathbb{C}$ with

$$(h+x)\cdot c = \lambda(h)c \qquad (h \in \mathfrak{h},\ x \in \mathfrak{n}_+,\ c \in \mathbb{C}).$$

Write $\chi(y)$ for the scalar by which $y \in \mathfrak{b}$ acts. Scalars commute, and χ vanishes on $\mathfrak{n}_+ \supseteq [\mathfrak{b},\mathfrak{b}]$. Hence

$$\chi([y,z]) = 0 = [\chi(y),\chi(z)],$$

and $\mathbb{C}_\lambda$ is a $\mathfrak{b}$-module. The map χ extends to an algebra homomorphism $\chi_\lambda : U(\mathfrak{b}) \to \mathbb{C}$.

Definition 12.3 (Verma module). The *Verma module* of highest weight λ is

$$M(\lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda,$$

with $U(\mathfrak{g})$ acting on the left factor. Put $v_\lambda = 1 \otimes 1$.

This construction parallels the induced representation $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ of the finite-group course, Lecture 8. Here $U(\mathfrak{g})$ replaces the group algebra $\mathbb{C}[G]$, and $U(\mathfrak{b})$ replaces the subgroup algebra $\mathbb{C}[H]$.

In this tensor product,

$$ub \otimes c = u \otimes \chi_\lambda(b)c \qquad (u \in U(\mathfrak{g}),\ b \in U(\mathfrak{b}),\ c \in \mathbb{C}_\lambda).$$

Taking $u = 1$ gives

$$xv_\lambda = 0 \quad (x \in \mathfrak{n}_+), \qquad hv_\lambda = \lambda(h)v_\lambda \quad (h \in \mathfrak{h}).$$

These relations do not show $v_\lambda \neq 0$. Proposition 12.4 below proves it from PBW.

Order the positive roots as $\beta_1, \dots, \beta_N$ with $\beta_i = \alpha_i$ for $i \le r$. Choose nonzero $f_j \in \mathfrak{g}_{-\beta_j}$, taking f_i from import (a) when $i \le r$. For $k \in \mathbb{Z}^N_{\ge 0}$ put

$$f^k = f_1^{k_1} f_2^{k_2} \cdots f_N^{k_N}, \qquad \nu(k) = \sum_{j=1}^{N} k_j \beta_j \in Q_+.$$

The *Kostant partition function* $\mathcal{P}(\nu)$ is the number of k with $\nu(k) = \nu$. The height $\operatorname{ht}\bigl(\sum_i c_i\alpha_i\bigr) = \sum_i c_i$, with $c_i \in \mathbb{Z}$, is well defined on the root lattice $Q = \sum_{i=1}^{r} \mathbb{Z}\alpha_i$, because the simple roots are linearly independent by import (b). Each $\beta_j \in Q_+ \setminus \{0\}$ has height at least 1. So $\nu(k) = \nu$ forces

$$k_1 + \cdots + k_N \le \sum_{j=1}^{N} k_j \operatorname{ht}\beta_j = \operatorname{ht}\nu.$$

Hence $\mathcal{P}(\nu)$ is finite. In particular $\mathcal{P}(0) = 1$.

Proposition 12.4 (Structure of $M(\lambda)$)**.** *The vectors $f^k v_\lambda$, $k \in \mathbb{Z}^N_{\ge 0}$, form a basis of $M(\lambda)$, and $f^k v_\lambda \in M(\lambda)_{\lambda - \nu(k)}$. Consequently:*

(i) *$v_\lambda \neq 0$, and $y \mapsto yv_\lambda$ is a linear isomorphism $U(\mathfrak{n}_-) \to M(\lambda)$;*

(ii) *$M(\lambda) = \bigoplus_{\nu \in Q_+} M(\lambda)_{\lambda-\nu}$ and $\dim M(\lambda)_{\lambda-\nu} = \mathcal{P}(\nu) < \infty$;*

(iii) *$M(\lambda)_\lambda = \mathbb{C}v_\lambda$.*

Proof. Step 1: freeness. The multiplication isomorphism of import (c) commutes with right multiplication by $U(\mathfrak{b})$. Hence, as right $U(\mathfrak{b})$-modules, $U(\mathfrak{g}) \cong U(\mathfrak{n}_-) \otimes U(\mathfrak{b})$, and

$$\begin{aligned} M(\lambda) &\cong \bigl(U(\mathfrak{n}_-) \otimes U(\mathfrak{b})\bigr) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda \\ &\cong U(\mathfrak{n}_-) \otimes \bigl(U(\mathfrak{b}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda\bigr) \\ &\cong U(\mathfrak{n}_-) \otimes \mathbb{C}_\lambda. \end{aligned}$$

Following $y \otimes 1$ back through these maps gives yv_λ. So $y \mapsto yv_\lambda$ is a linear isomorphism. The f^k form a basis of $U(\mathfrak{n}_-)$ by PBW for $\mathfrak{n}_-$. This proves the basis statement and (i).

Step 2: weights. For $h \in \mathfrak{h}$, the commutator $[h, \cdot\,]$ on $U(\mathfrak{g})$ satisfies

$$[h, yz] = [h, y]z + y[h, z].$$

Since $[h, f_j] = -\beta_j(h) f_j$, induction on the length of f^k gives

$$[h, f^k] = -\nu(k)(h)\, f^k.$$

Therefore

$$hf^k v_\lambda = f^k h v_\lambda + [h, f^k]v_\lambda = \lambda(h)\, f^k v_\lambda - \nu(k)(h)\, f^k v_\lambda = \bigl(\lambda - \nu(k)\bigr)(h)\, f^k v_\lambda.$$

Step 3: weight spaces. Let B_ν be the span of the $f^k v_\lambda$ with $\nu(k) = \nu$. Steps 1 and 2 give

$$M(\lambda) = \bigoplus_{\nu \in Q_+} B_\nu, \qquad B_\nu \subseteq M(\lambda)_{\lambda - \nu}, \qquad \dim B_\nu = \mathcal{P}(\nu).$$

Let $w \in M(\lambda)_\mu$ and write $w = \sum_\nu w_\nu$ with $w_\nu \in B_\nu$. The line $\mathbb{C}w$ is stable under $\mathfrak{h}$, so Lemma 12.1 gives $w_\nu \in \mathbb{C}w \subseteq M(\lambda)_\mu$. If $\lambda - \nu \neq \mu$, then $w_\nu \in M(\lambda)_\mu \cap M(\lambda)_{\lambda-\nu} = 0$ by directness of weight spaces. So $M(\lambda)_{\lambda-\nu} = B_\nu$, and there are no other weights. This proves (ii). Part (iii) is the case $\nu = 0$. ∎

For $\mathfrak{sl}_3$ the positive roots are α_1, α_2 and $\alpha_1 + \alpha_2$. Exercise 12.3 proves that $\mathcal{P}(a\alpha_1 + b\alpha_2) = \min(a, b) + 1$. Figure 12.1 shows these weight multiplicities of $M(\lambda)$.

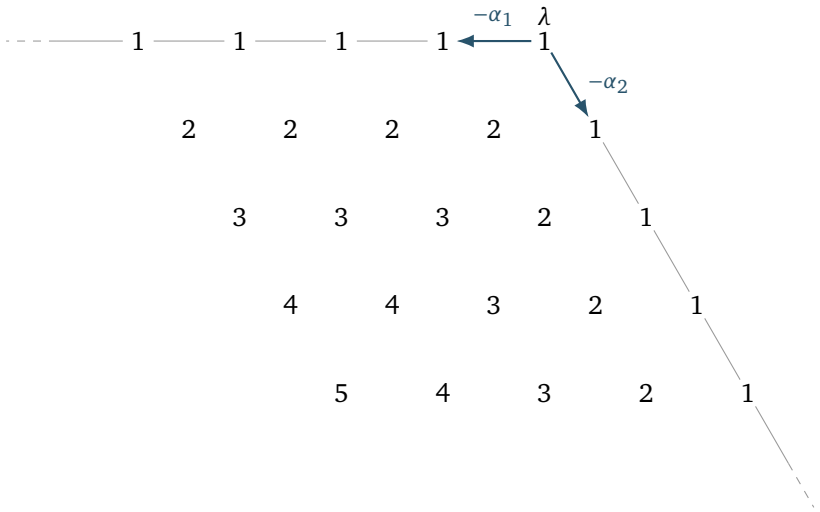


Figure 12.1: Weight multiplicities of the $\mathfrak{sl}_3$ Verma module $M(\lambda)$. The number at $\lambda - a\alpha_1 - b\alpha_2$ is $\mathcal{P}(a\alpha_1 + b\alpha_2)$, drawn for $0 \le a, b \le 4$. The cone continues past the drawn region.

12.2 Highest-weight modules and the universal property

Definition 12.5 (Highest vector and highest-weight module)**.** Let V be a $\mathfrak{g}$-module. A *highest vector* of weight λ is a nonzero $v \in V_\lambda$ with $\mathfrak{n}_+ v = 0$. The module V is a *highest-weight module* of highest weight λ if $V = U(\mathfrak{g})v$ for such a vector v.

By Proposition 12.4, v_λ is a highest vector and $M(\lambda)$ is a highest-weight module.

Proposition 12.6 (Universal property)**.** *Let V be a $\mathfrak{g}$-module and $v \in V$ with $\mathfrak{n}_+ v = 0$ and $hv = \lambda(h)v$ for all $h \in \mathfrak{h}$. There is a unique module map $\varphi \colon M(\lambda) \to V$ with $\varphi(v_\lambda) = v$. Its image is $U(\mathfrak{g})v$.*

Proof. The set of $b \in U(\mathfrak{b})$ with $bv = \chi_\lambda(b)v$ is a subalgebra containing 1 and $\mathfrak{b}$. Since $\mathfrak{b}$ generates the algebra $U(\mathfrak{b})$, this subalgebra is all of $U(\mathfrak{b})$.

Define $\Phi(u, c) = c\,uv$ for $u \in U(\mathfrak{g})$ and $c \in \mathbb{C}_\lambda$. It is bilinear and balanced over $U(\mathfrak{b})$:

$$\Phi(ub, c) = c\,ubv = c\,\chi_\lambda(b)\,uv = \Phi\big(u, \chi_\lambda(b)c\big).$$

So it induces a linear map

$$\varphi\colon M(\lambda) \longrightarrow V, \qquad \varphi(u \otimes c) = c\,uv.$$

For $u' \in U(\mathfrak{g})$,

$$\varphi\big(u'(u \otimes c)\big) = \varphi(u'u \otimes c) = c\,u'uv = u'\varphi(u \otimes c).$$

Thus φ is a module map with image $U(\mathfrak{g})v$. It is unique because $\varphi(uv_\lambda) = u\varphi(v_\lambda)$ and v_λ generates $M(\lambda)$. ∎

Thus every highest-weight module of highest weight λ is a quotient of $M(\lambda)$.

Corollary 12.7 (Shape of a highest-weight module). *Let V be a highest-weight module with highest vector v of weight λ. Then*

$$V = U(\mathfrak{n}_-)v, \qquad V = \bigoplus_{\nu \in Q_+} V_{\lambda-\nu}, \qquad \dim V_{\lambda-\nu} \le \mathcal{P}(\nu), \qquad V_\lambda = \mathbb{C}v.$$

Proof. Let $\varphi\colon M(\lambda) \to V$ be the surjection given by Proposition 12.6 with $\varphi(v_\lambda) = v$. The map φ commutes with $\mathfrak{h}$, so $\varphi(M(\lambda)_{\lambda-\nu}) \subseteq V_{\lambda-\nu}$. By surjectivity and Proposition 12.4(ii),

$$V = \sum_{\nu \in Q_+} \varphi\big(M(\lambda)_{\lambda-\nu}\big).$$

This sum is direct by Lemma 12.1. Comparing weight components as in Step 3 of the proof of Proposition 12.4 gives $V_{\lambda-\nu} = \varphi(M(\lambda)_{\lambda-\nu})$, and no other weights. The dimension bound follows. The case $\nu = 0$ gives $V_\lambda = \mathbb{C}v$, and $\varphi(U(\mathfrak{n}_-)v_\lambda) = U(\mathfrak{n}_-)v$ gives the first equality. ∎

12.3 The maximal submodule and $L(\lambda)$

Theorem 12.8 (Unique maximal proper submodule). *Let V be a highest-weight module with highest vector v of weight λ. Put*

$$V' = \bigoplus_{\nu \in Q_+ \setminus \{0\}} V_{\lambda-\nu}.$$

(i) A submodule $W \subseteq V$ is proper if and only if $W \subseteq V'$.

(ii) The sum N_V of all proper submodules is proper. It is the unique maximal proper submodule.

(iii) V/N_V is simple, and it is the only simple quotient of V.

Proof. (i) Let W be a submodule with $W \not\subseteq V'$. Choose $w \in W \setminus V'$ and write

$$w = w_0 + w', \qquad w_0 \in V_\lambda \setminus \{0\}, \quad w' \in V'.$$

The submodule W is stable under $\mathfrak{h}$, so Lemma 12.1 gives $w_0 \in W$. Since $V_\lambda = \mathbb{C}v$,

$$w_0 = av \quad (a \neq 0), \qquad v = a^{-1}w_0 \in W.$$

Hence

$$W \supseteq U(\mathfrak{g})v = V.$$

Conversely, if $W \subseteq V'$, then $v \notin W$ because $V_\lambda \cap V' = 0$. So W is proper.

(ii) Every proper submodule lies in V'. So their sum lies in V', and it is proper by (i). It contains every proper submodule.

(iii) Submodules of V/N_V correspond to submodules of V containing N_V. By (ii) these are N_V and V. Since $V/N_V \neq 0$, it is simple. If V/K is simple, the same correspondence makes K a maximal proper submodule. By (ii), $K = N_V$. ∎

Definition 12.9 (The module $L(\lambda)$). Let $N(\lambda)$ be the maximal proper submodule of $M(\lambda)$. Define

$$L(\lambda) = M(\lambda)/N(\lambda).$$

Corollary 12.10 (Simple highest-weight modules). *(i) $L(\lambda)$ is a simple highest-weight module of highest weight λ.*

(ii) Every simple highest-weight module of highest weight λ is isomorphic to $L(\lambda)$.

(iii) If $L(\lambda) \cong L(\mu)$, then $\lambda = \mu$.

Proof. (i) Simplicity is Theorem 12.8(iii). The image $\bar{v}_\lambda$ of v_λ generates $L(\lambda)$. Write $M(\lambda)'$ for the subspace V' of Theorem 12.8 with $V = M(\lambda)$. The vector $\bar{v}_\lambda$ is nonzero because $N(\lambda) \subseteq M(\lambda)'$ does not contain v_λ. It is a highest vector of weight λ.

(ii) Let V be such a module. Proposition 12.6 gives a surjection $M(\lambda) \to V$ with kernel K. Since $M(\lambda)/K \cong V$ is simple, Theorem 12.8(iii) gives $K = N(\lambda)$.

(iii) By Corollary 12.7, λ is a weight of $L(\lambda)$ and every weight of $L(\lambda)$ is $\leq \lambda$. An isomorphism preserves weights. Hence $\lambda \leq \mu$ and $\mu \leq \lambda$, and import (b) gives $\lambda = \mu$. ∎

12.4 Finite-dimensional simple modules

Lemma 12.11 (A rank-one computation). *Let e, h, f act on a module V and satisfy $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. If $v \in V$, $ev = 0$ and $hv = cv$, then for $k \geq 0$*

$$hf^k v = (c - 2k)f^k v, \qquad ef^{k+1}v = (k+1)(c-k)f^k v.$$

Proof. The first formula follows by induction from $hf = fh - 2f$:

$$hf^{k+1}v = fhf^k v - 2f^{k+1}v = (c-2k)f^{k+1}v - 2f^{k+1}v = \big(c - 2(k+1)\big)f^{k+1}v.$$

For the second, the case $k = 0$ is

$$efv = [e, f]v + fev = hv = cv.$$

Assume the case $k - 1$. Then

$$\begin{aligned} ef^{k+1}v &= [e, f]f^k v + f\, ef^k v \\ &= (c - 2k)f^k v + k(c - k + 1)f^k v \\ &= \big((k+1)c - k^2 - k\big)f^k v \\ &= (k+1)(c-k)f^k v. \end{aligned}$$ ∎

Theorem 12.12 (Finite-dimensional simple modules). *Let V be a finite-dimensional simple $\mathfrak{g}$-module. Then V contains a highest vector of some weight λ, with*

$$\lambda(h_i) \in \mathbb{Z}_{\geq 0} \qquad (i = 1, \ldots, r).$$

Thus $\lambda \in P_+$ and $V \cong L(\lambda)$. The weight λ is determined by V.

Proof. Step 1: a weight exists. Let $t_1, \dots, t_r$ be a basis of $\mathfrak{h}$. Write $\rho(x)$ for the operator of $x \in \mathfrak{g}$ on V. The operators $\rho(t_j)$ commute, since $\mathfrak{h}$ is abelian. An eigenspace of $\rho(t_1)$ is stable under each $\rho(t_j)$. Repeating inside it, we find a common eigenvector. Its eigenvalues define a weight. By Lemma 12.1, V has at most $\dim V$ weights.

Step 2: a highest vector. Choose a weight λ that is maximal for $\leq$ in this finite set. For $\alpha \in R_+$, the weight $\lambda + \alpha > \lambda$ does not occur, so

$$\mathfrak{g}_\alpha V_\lambda \subseteq V_{\lambda+\alpha} = 0.$$

Thus every nonzero $v \in V_\lambda$ is a highest vector. The submodule $U(\mathfrak{g})v$ is nonzero, so it equals V. Corollary 12.10(ii) gives $V \cong L(\lambda)$, and (iii) shows that λ is determined by V.

Step 3: dominance and integrality. Fix i and put $c = \lambda(h_i)$. Since $e_i \in \mathfrak{g}_{\alpha_i} \subseteq \mathfrak{n}_+$ and $v \in V_\lambda$,

$$e_i v = 0, \qquad h_i v = cv.$$

The triple (e_i, h_i, f_i) satisfies the relations of import (a). Apply Lemma 12.11 to this triple and v. The nonzero vectors among the $f_i^k v$ have distinct h_i-eigenvalues $c - 2k$, so they are linearly independent. Since $\dim V < \infty$, some $f_i^k v$ is zero. Let k be least with $f_i^k v = 0$. Since $v \neq 0$, we have $k \geq 1$ and $f_i^{k-1} v \neq 0$. The second formula of Lemma 12.11 gives

$$0 = e_i f_i^k v = k(c - k + 1) f_i^{k-1} v.$$

Hence

$$\lambda(h_i) = c = k - 1 \in \mathbb{Z}_{\geq 0}.$$ ∎

Remark 12.13 (Lie's theorem gives a highest vector)**.** This remark gives another way to obtain the highest vector of Step 2. Nothing later depends on it. Let V be a nonzero finite-dimensional $\mathfrak{g}$-module.

First, the algebra $\mathfrak{b}$ is solvable and $[\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}_+$. Brackets of root vectors add heights by import (a), and the positive roots have bounded height, so the derived series of $\mathfrak{n}_+$ reaches 0. We saw above that $[\mathfrak{b}, \mathfrak{b}] \subseteq \mathfrak{n}_+$, so $\mathfrak{b}$ is solvable. Conversely, let $x \in \mathfrak{g}_\alpha$ with $\alpha \in R_+$, and choose $h \in \mathfrak{h}$ with $\alpha(h) \neq 0$. Then $x = \alpha(h)^{-1}[h, x]$, so $\mathfrak{n}_+ \subseteq [\mathfrak{b}, \mathfrak{b}]$.

Next, Lie's theorem in Lecture 3 gives a nonzero $v \in V$ and a linear map $\eta \colon \mathfrak{b} \to \mathbb{C}$ with $yv = \eta(y)v$ for all $y \in \mathfrak{b}$. The same theorem says that η vanishes on $[\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}_+$. So $\mathfrak{n}_+ v = 0$, and $\mathfrak{h}$ acts on v by a weight. Thus v is a highest vector. If V is simple, then v generates V.

This argument uses neither complete reducibility nor the classification. It does not replace Step 3, which proves integrality.

12.5 Worked example: Verma modules for $\mathfrak{sl}_2$

This section returns to the strings of Exercise 6.7, which do not stop. They are the Verma modules of $\mathfrak{sl}_2$.

Let $\mathfrak{g} = \mathfrak{sl}_2$ with basis (e, h, f). Then $\mathfrak{h} = \mathbb{C}h$, $\mathfrak{n}_+ = \mathbb{C}e$ and $\mathfrak{n}_- = \mathbb{C}f$. The positive root α has $\alpha(h) = 2$. Identify λ with $c = \lambda(h) \in \mathbb{C}$, and write $M(c)$ and $L(c)$. Write $v_0 = v_\lambda$ for the generator of $M(c)$. Since $\mathcal{P}(k\alpha) = 1$, the vectors

$$v_k = f^k v_0 \qquad (k \geq 0)$$

form a basis of $M(c)$, one in each weight $c - 2k$. Lemma 12.11 gives the action, with $v_{-1} = 0$:

$$hv_k = (c - 2k)v_k, \qquad fv_k = v_{k+1}, \qquad ev_k = k(c - k + 1)v_{k-1}.$$

Iterating the last formula, for $k \geq 1$,

$$\begin{aligned} e^k v_k &= k(c-k+1)\, e^{k-1} v_{k-1} \\ &= \prod_{j=1}^{k} j(c-j+1)\; v_0. \end{aligned}$$

Highest vectors. A weight vector killed by e is a multiple of some v_k. For $k \geq 1$,

$$ev_k = 0 \iff k(c-k+1) = 0 \iff c = k-1.$$

So a highest vector outside $\mathbb{C}v_0$ exists exactly when $c = m \in \mathbb{Z}_{\geq 0}$. It is $v_{m+1} = f^{m+1} v_0$, up to a scalar.

Generic c. Let $c \notin \mathbb{Z}_{\geq 0}$ and let $W \neq 0$ be a submodule. Each weight space is a line $\mathbb{C}v_k$, so Lemma 12.1 gives some $v_k \in W$. Every factor $j(c-j+1)$ is nonzero, so $e^k v_k$ is a nonzero multiple of v_0. Hence $v_0 \in W$ and $W = M(c)$. Thus $M(c) = L(c)$ is simple and infinite-dimensional.

Dominant integral $c = m$. The vector v_{m+1} is a highest vector:

$$\begin{aligned} ev_{m+1} &= (m+1)(m-m)\, v_m = 0, \\ hv_{m+1} &= \big(m - 2(m+1)\big) v_{m+1} = (-m-2)\, v_{m+1}. \end{aligned}$$

By Corollary 12.7,

$$U(\mathfrak{g}) v_{m+1} = U(\mathfrak{n}_-) v_{m+1} = \operatorname{span}\{v_k : k \geq m+1\}.$$

This submodule lies in $M(m)' = \operatorname{span}\{v_k : k \geq 1\}$, so it is proper by Theorem 12.8(i). We show that it contains every proper submodule, so that it equals $N(m)$. Let W be a proper submodule. By Theorem 12.8(i) and Lemma 12.1, W is spanned by some v_k with $k \geq 1$. If $v_k \in W$ with $1 \leq k \leq m$, then

$$e^k v_k = \prod_{j=1}^{k} j(m-j+1)\; v_0 = k! \, \frac{m!}{(m-k)!}\, v_0 \neq 0,$$

so $v_0 \in W$ and $W = M(m)$, which contradicts properness. Hence

$$N(m) = \operatorname{span}\{v_k : k \geq m+1\}, \qquad \dim L(m) = m+1.$$

Figure 12.2 shows the case $m = 2$. Exercise 12.1 proves $N(m) \cong M(-m-2)$.

The images $\bar{v}_0, \dots, \bar{v}_m$ form a basis of $L(m)$. Their action formulas are those of V_m in Lecture 6, so $L(m) \cong V_m$. Since $\omega_1(h) = 1$, we write $V_m = L(m\omega_1)$.

Highest weight versus finite dimension. The module $M(m)$ is infinite-dimensional, although its highest weight is dominant integral. Its nonzero submodule $N(m)$ has no complement. A complementary submodule C would be proper. The maximal-submodule theorem gives $C \subseteq N(m)$, so $C = C \cap N(m) = 0$. This contradicts $M(m) = N(m) \oplus C$. Weyl's theorem in Lecture 5 concerns finite-dimensional modules. Here the simple quotient $L(m)$ is finite-dimensional.

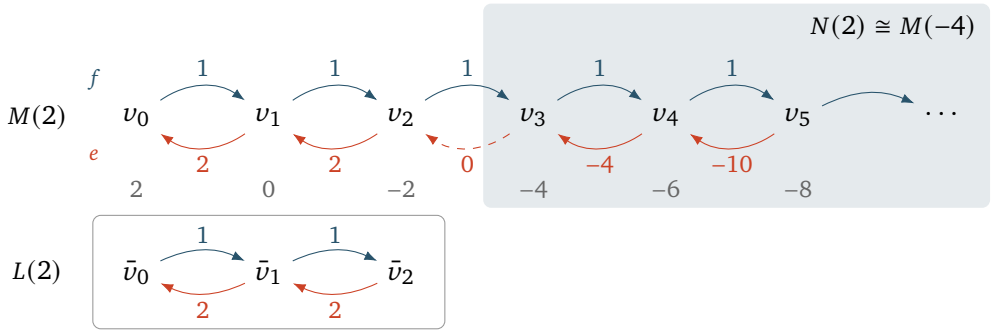


Figure 12.2: The Verma module $M(2)$, with the weight of each v_k below it. Upper arrows are f and lower arrows are e, labelled with their coefficients. The coefficient of e from v_3 to v_2 is 0, so the shaded tail is a submodule, while f still enters it from v_2. The box holds the images of v_0, v_1, v_2 in the quotient $L(2)$.

Self-checks

(1) For $\mathfrak{g} = \mathfrak{sl}_3$ with simple roots α_1, α_2, use PBW to give a basis of $M(\lambda)_{\lambda-\alpha_1-\alpha_2}$.
(2) Is the $\mathfrak{sl}_2$ Verma module $M(-1)$ simple?
(3) Can a finite-dimensional $\mathfrak{sl}_2$-module contain $v \neq 0$ with $ev = 0$ and $hv = -2v$?

Answers. (1) A basis is $f_1 f_2 v_\lambda$ and $f_3 v_\lambda$, where $f_3 \in \mathfrak{g}_{-\alpha_1-\alpha_2}$. So the dimension is two. (2) Yes. Here $c = -1 \notin \mathbb{Z}_{\geq 0}$. For $k \geq 1$, the coefficients $k(c-k+1)$ are $k(-k) = -k^2 \neq 0$. (3) No. Step 3 of the proof of Theorem 12.12 uses only $v \neq 0$, $ev = 0$, $hv = cv$ and $\dim V < \infty$. With $c = -2$, it would give $-2 \in \mathbb{Z}_{\geq 0}$.

Summary

PBW makes $M(\lambda)$ free over $U(\mathfrak{n}_-)$, with weight spaces of dimension $\mathcal{P}(\nu)$ and a one-dimensional top weight space.

Every highest-weight module of highest weight λ is a quotient of $M(\lambda)$. The weight decomposition gives a unique maximal proper submodule, hence a unique simple quotient $L(\lambda)$. A finite-dimensional simple module is isomorphic to $L(\lambda)$ for a unique $\lambda \in P_+$. Lecture 13 proves the converse, using the vectors $f_i^{m_i+1} v_\lambda$ with $m_i = \lambda(h_i)$, seen here for $\mathfrak{sl}_2$.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 20 and Subsection 21.1. Kirillov, *An Introduction to Lie Groups and Lie Algebras*, Sections 8.2–8.3.